

Pressure in Solids and Liquids — Paper D

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a **unit**. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- No formulae and no conversion factors are given on this paper.
- Several questions describe situations you may not have seen before — apply the ideas you already have.
- Take **$g = 10 \text{ N/kg}$** throughout, and give pressures in pascals unless told otherwise.

SECTION 1 – DESIGNING FOR PRESSURE

- 1.** An elephant of weight 40 000 N stands on four feet, each of area 0.1 m^2 . A woman of weight 500 N stands on a single stiletto heel of area 0.0001 m^2 .

(a) Calculate the pressure the elephant exerts on the ground. [2]

(b) Calculate the pressure under the heel. [2]

(c) Explain how the far lighter woman can exert the far greater pressure. [2]

- 2.** A sledge is to carry a load of 1800 N across snow. The snow gives way if the pressure on it exceeds 4000 Pa.

- (a)** Calculate the smallest total runner area that will keep the sledge on top of the snow. [3]

- (b)** The sledge has two runners of equal area. State the area of each. [1]

- (c)** The load is increased to 2250 N with the same runners. Show that the sledge now sinks. [1]

SECTION 2 – LIQUIDS THAT IGNORE THE CONTAINER

- 3.** A tank is narrow at the top and much wider at the bottom. It is filled with water of density 1000 kg/m^3 to a depth of 2.5 m.

- (a)** Calculate the pressure due to the water at the bottom of the tank. [2]

- (b)** The tank is replaced by a straight-sided tank half as wide, filled to the same depth. State the pressure at the bottom now, and justify your answer. [2]

- (c)** State the directions in which this pressure acts. [1]

- 4.** A glass tube is filled with mercury, of density $13\,600 \text{ kg/m}^3$.

- (a)** Calculate the pressure produced by a column of mercury 0.75 m tall. [2]

- (b)** Atmospheric pressure is about 100 000 Pa. Explain why a mercury column of roughly this height is used to measure it. [2]

SECTION 3 – JUDGEMENT

- 5.** Two identical tanks are filled to a depth of 5 m, one with water (1000 kg/m^3) and one with cooking oil (800 kg/m^3).

- (a)** Calculate the pressure at the bottom of each tank. [2]

- (b)** Calculate the depth of oil that would give the same pressure as 5 m of water. [2]

6. A student has written two answers, and both are wrong.

[6]

(i) “A force of 60 N on an area of 20 cm² gives a pressure of 3 Pa.”

(ii) “Two tanks are both filled with water to a depth of 2 m. The pressure at the bottom of the wide tank is greater than at the bottom of the narrow one, because the wide tank holds more water.”

For each one, explain the mistake and give the correct answer.

End of paper. Check every answer has a unit or a form the question actually asked for.